PeopleDesk HRMS

Enterprise SaaS Human Resource Management Platform

Technical Architecture Overview

Prepared for Enterprise Architecture Review

Customer Confidential · Version 2.1 · 25 July 2026

Why PeopleDesk HRMS

Six positions, each grounded in the production implementation

Position	Basis
Enterprise Ready	19 business domains · 600+ permission controls · 40+ approval workflows
SaaS Native	Tenant-partitioned with a three-level organizational hierarchy
Production Proven	120+ live customers at multi-million-row scale
Cloud Deployable	Containerized, orchestrated, configuration injected per environment
Open and Integrable	1,500+ documented REST endpoints plus device connectivity
Layered Security Controls	Identity, authorization, transport, data and audit

Platform at a Glance

Scale of the deployed platform

Dimension	Measure
Business domains	19
REST endpoints	1,500+
Application screens	750+
Standard reports	130+
Data platform	570+ tables · 18 schemas
Permission controls	600+
Automated processes	45+
Customers	120+

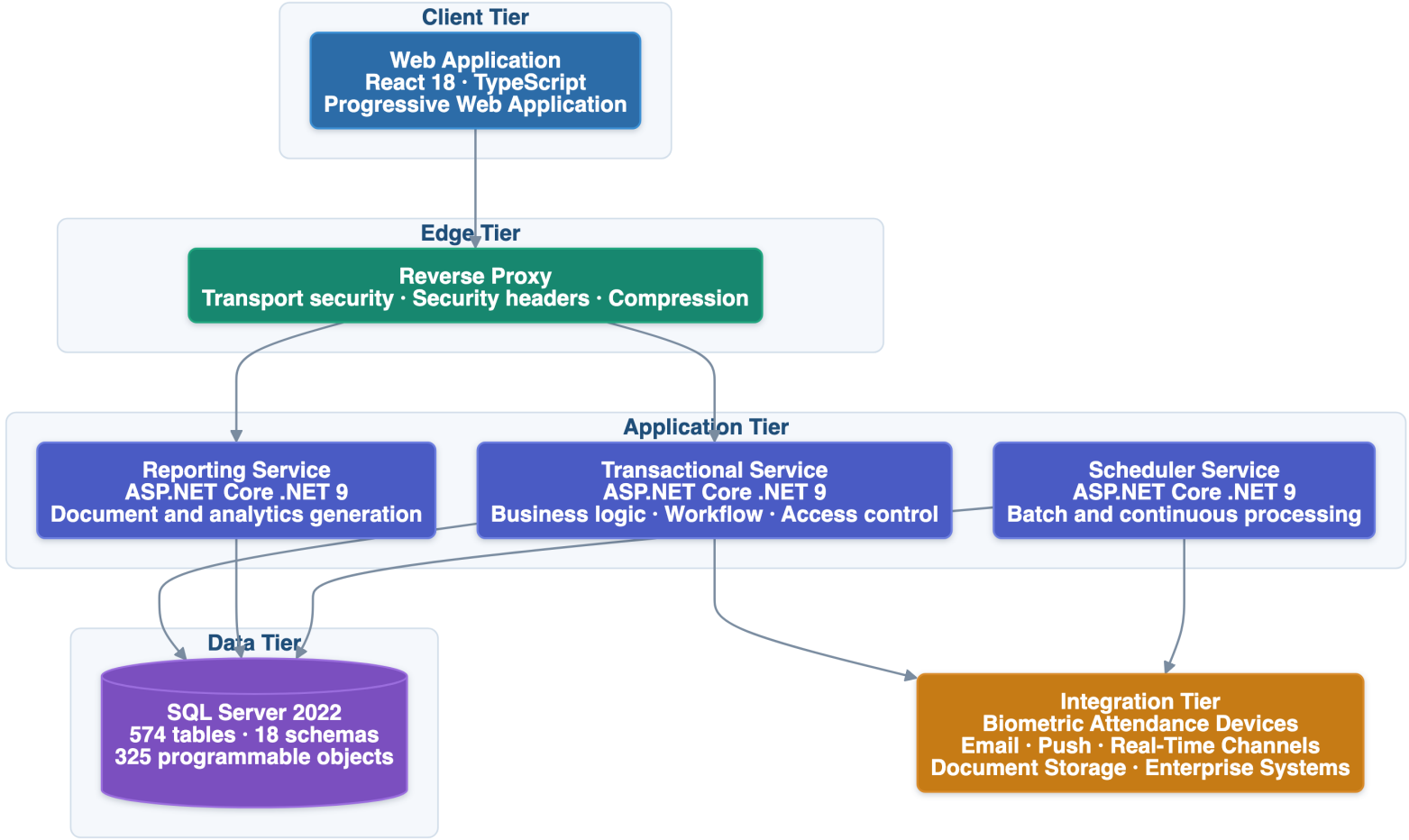
Product Overview

Nineteen functional domains on one platform and one permission model

- **Core HR —** Employee Management, Organizational Framework, Configuration
- **Time & Attendance —** Attendance, Leave & Movement, Roster & Shift
- **Payroll & Benefits —** Payroll & Tax, Allowances, Loan, Benefits, Retirement
- **Talent —** Recruitment, Performance, Training, Manpower Analysis
- **Governance —** Approval Workflow, Asset, Expense, Employee Self-Service
- **Depth —** 600+ permission controls · 40+ configurable approval workflows

Solution Architecture

Four independently deployable services over a unified data platform — interactive, reporting and batch workloads physically separated



Architecture Principles

Seven principles applied consistently across the platform

Principle	Application in PeopleDesk HRMS
Separation of Concerns	Workloads isolated by service boundary
Stateless Services	Token-based identity, no server-side session state
Configuration over Customization	Structure, roles and workflows configured per tenant
Open API Surface	Capability exposed through documented REST interfaces
Layered Security	Controls at identity, authorization, transport, data and audit
Cloud Ready	Containerized services with per-service orchestration manifests
Evidence-Based Evolution	Every change impact-analyzed against a knowledge model

Why This Architecture Matters

Architectural decisions translated into business outcomes

Decision	Business Outcome
Independent service deployment	Faster feature delivery
Workload-class separation	Lower operational risk
Stateless application tier	Enterprise scalability
Modular boundaries	Easier maintenance
Dedicated reporting tier	Predictable performance
Scheduler isolation	High-availability foundation
Configuration-led tenancy	Lower onboarding cost

Quality Attributes

Non-functional requirements and the design response

Quality Attribute	Design Response
Scalability	Stateless application services
Availability	Independently deployable services
Security	Layered security controls
Maintainability	Modular architecture
Extensibility	Service separation
Performance	Dedicated reporting service
Reliability	Background scheduler isolation

Technology Stack

Current-generation technology across every tier

Layer	Technology
Frontend	React 18 · TypeScript · Vite 7 · Redux Toolkit
Backend	ASP.NET Core .NET 9 · MediatR · FluentValidation
Data Access	EF Core 9 (writes) · Dapper (reads) · Stored procedures
Database	SQL Server 2022
Authentication	JWT bearer · refresh-token rotation · scope claims
Reporting	RDLC · Handlebars + headless rendering · ClosedXML
Background	Quartz.NET · .NET Hosted Services
Infrastructure	Docker · Reverse proxy · Kubernetes

Selection criteria: modern · stable · long-term supported · enterprise proven · cloud compatible

Security Architecture

Layered controls across five domains

- **Authentication —** JWT bearer with issuer and signing-key validation; server-side refresh-token rotation; federated sign-in
- **Authorization —** 600+ permission controls; role-based assignment; organizational-scope claims in the security token
- **Application data —** payload protection, declarative validation, centralized exception handling, output sanitization
- **Transport & edge —** HTTPS enforcement, HSTS, frame protection, content-type enforcement, browser feature restriction
- **Audit —** request-level audit trail; background process execution history

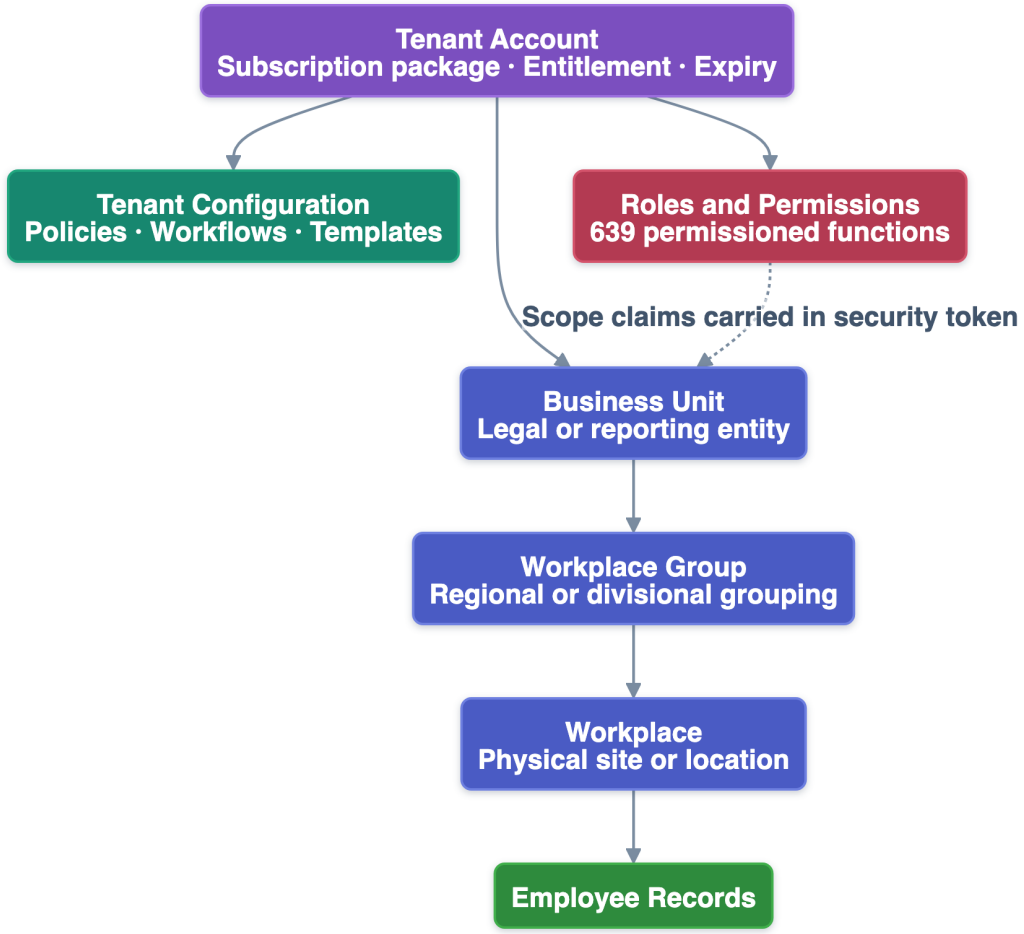
Secure Development Practices

Security controls mapped to the OWASP Secure Coding Practices categories

OWASP Category	Implemented Controls
Input Validation	Server-side validation pipeline on every command
Output Encoding	Untrusted markup neutralized before rendering
Session Management	Token validation with server-side refresh-token rotation
Access Control	600+ permissioned functions with organizational scope
Error Handling & Logging	Centralized handling; request-level audit trail
Communication Security	HTTPS with HSTS and the full edge header set
Database Security	Parameterized queries and stored-procedure access
System Configuration	Current supported versions; per-environment configuration

SaaS Architecture

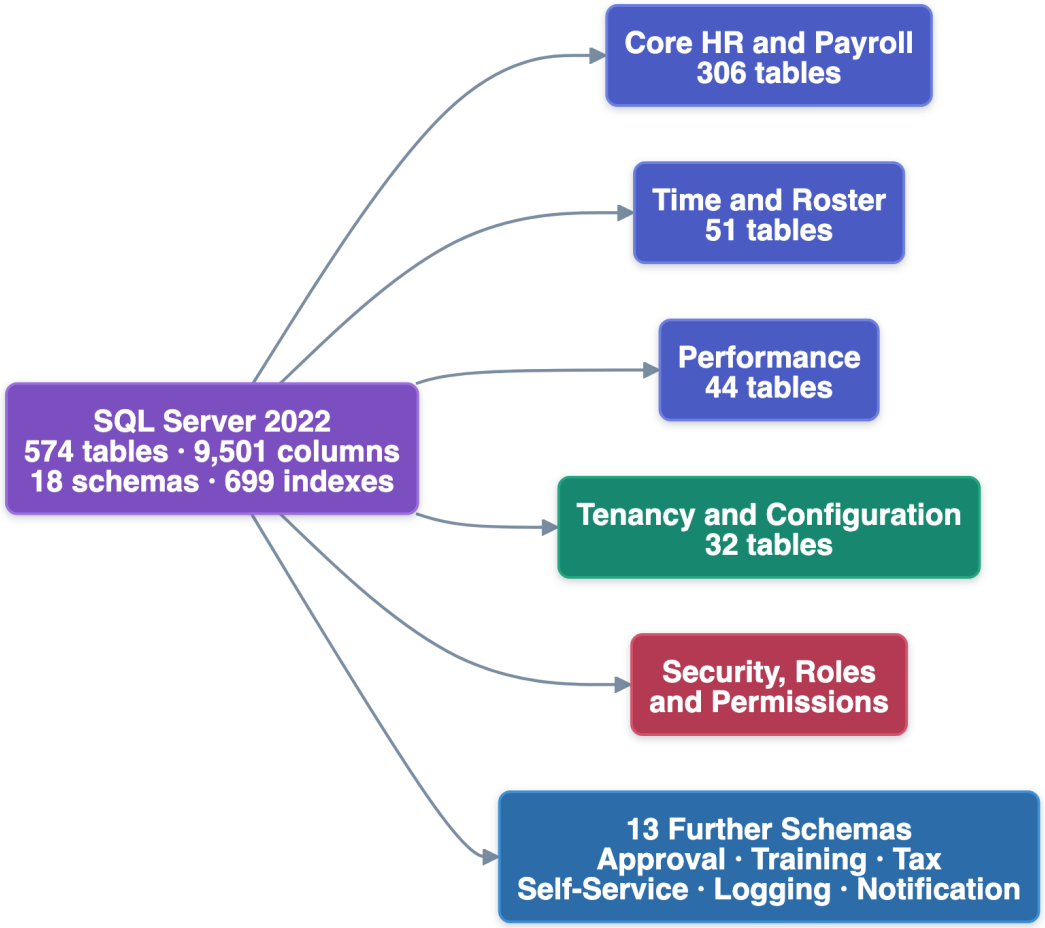
Tenant-partitioned with enterprise organizational depth



- Every business record keyed to its owning tenant account
- Three-level hierarchy — business unit → workplace group → workplace
- Organizational scope carried in the security token
- Automated tenant provisioning with seed configuration
- Basic 100 · Lite 500 · Premium 50,000 seats
- Seat and expiry entitlements enforced automatically

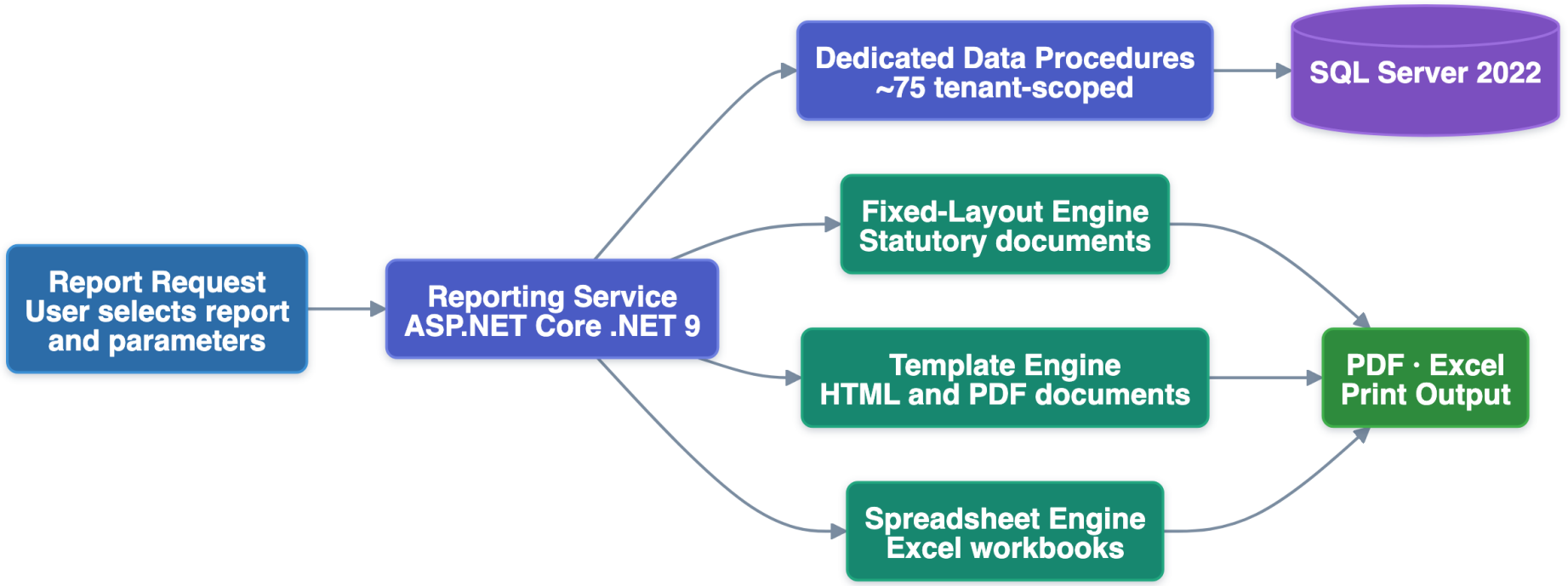
Data Architecture

570+ tables · 18 schemas · 690+ indexes — engineered for set-based processing at scale



Reporting & Analytics

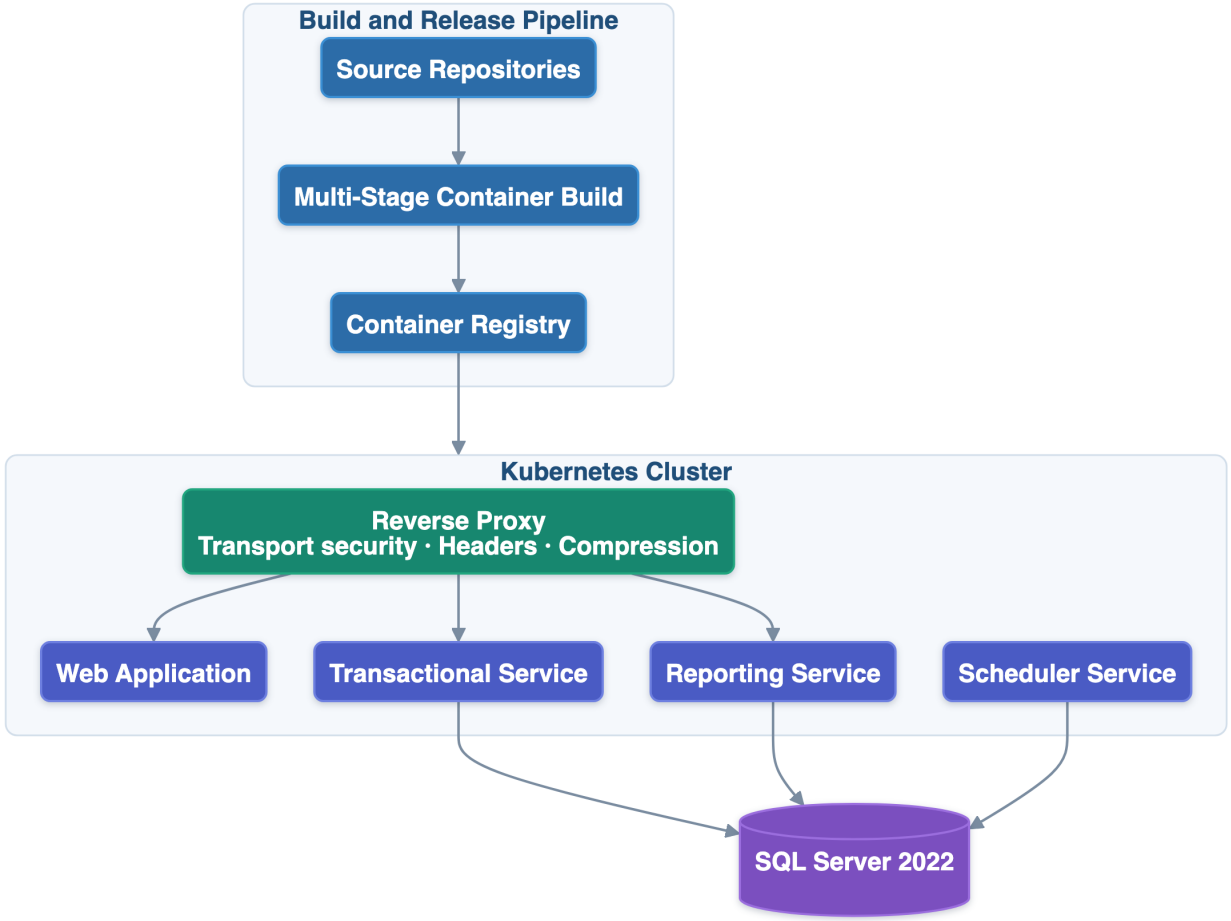
A separate reporting service — no contention with transactional workload



- 130+ standard reports served by dedicated tenant-scoped data procedures
- Render engine selected by output class — statutory, document, spreadsheet
- Organization-specific templates without forking the report catalogue

Deployment Architecture

Containerized, orchestrated, independently releasable



- One image promoted across environments; configuration injected per environment
- The artefact promoted to production is identical to the one validated in staging

Scalability & Performance

Scalability designed into the workload topology

- Batch payroll isolated from interactive workload by service boundary
- Stateless authentication removes application-tier affinity constraint
- Set-based computation at the data tier rather than application iteration
- 690+ indexes across a 570+ table production schema
- Route-level code splitting with vendor chunk separation on the frontend
- **Production evidence —** core operational tables sustain 20–40 million+ rows

Integration Ecosystem

Open interfaces and proven enterprise connectivity

- REST across 1,500+ endpoints with OpenAPI specification
- **Biometric attendance devices —** push-protocol capture plus 18+ scheduled collectors
- Real-time in-application notification channel
- Firebase Cloud Messaging push delivery
- SMTP email for notification and approval routing
- Azure Blob and FTP document storage · Managerium ERP integration

Knowledge-Driven Engineering

How the platform evolves safely over a multi-year dependency

- A twelve-layer structured knowledge base is the authoritative model of the system
- 160+ documents · 30+ decision tables · complete inter-module dependency graph
- Every proposed change is impact-analyzed against the model before implementation
- The first two phases of this framework are in production use today
- Addresses what architecture alone cannot: predictable evolution over time

Business Value

What the architecture delivers commercially

Benefit	Basis
Reduced implementation risk	Production-proven across 120+ enterprise customers
Lower operational cost	Configuration-led onboarding rather than code change
Independent scaling	Workload classes scale on separate boundaries
Future-ready platform	Current-generation runtimes, no end-of-life dependency
Enterprise governance	600+ permission controls with organizational scope
High maintainability	Modular boundaries plus a structured knowledge model

Technical Position Summary

PeopleDesk HRMS at a glance

Dimension	Position
Architecture	Four services, workload-class separation, unified data platform
Technology	.NET 9 · React 18 · SQL Server 2022 — no end-of-life dependency
Security	Layered controls mapped to the OWASP Secure Coding Practices
SaaS	Tenant-partitioned, three-level organizational hierarchy
Scale	Production-validated at multi-million-row operational volume
Integration	Open REST plus proven enterprise and device connectivity

Thank You

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